# GCSE PHYSICS

## Higher Tier – Paper 2 style

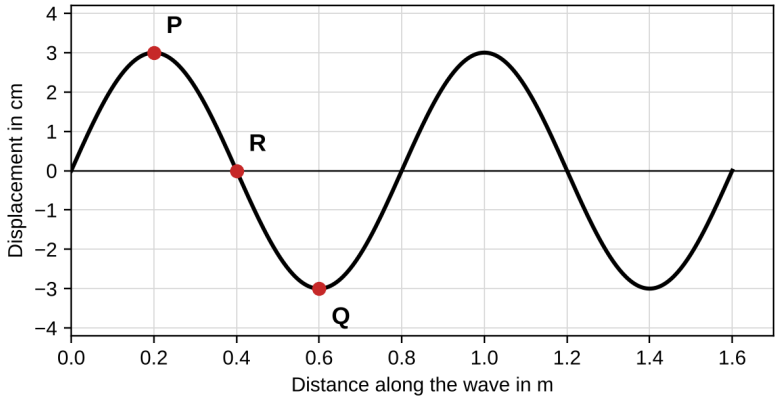
## Topic 4.6: Waves

*Practice questions in the style of AQA GCSE Physics (8463). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.*

| | |
|---|---|
| **Time allowed** | About 155 minutes (roughly 1 minute per mark) |
| **Materials** | For this paper you must have: a ruler, a pencil and a calculator. |
| **Instructions** | • Use black ink or a black ball-point pen (pencil for diagrams and graphs).<br>• Answer **all** questions.<br>• Answer the questions in the spaces provided.<br>• Do all rough work on a separate sheet.<br>• Cross through any work you do not want to be marked. |
| **Information** | • The maximum mark for this paper is **147**.<br>• The marks for questions are shown in brackets.<br>• You are allowed to use a calculator.<br>• You are reminded of the need for good English and clear presentation in your answers.<br>• Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence. |

Name: _____ Date: _____ Mark: _____ / 147

**0 1** The graph shows a transverse wave travelling along a rope at a certain moment. The wave speed is 2.0 m/s.




**0 1 . 1** Name the points **P**, **Q** and **R** on the wave. Choose from: crest, trough, rest position. **[3 marks]**

**0 1 . 2** Use the graph to find the amplitude of the wave. **[1 mark]**

**0 1 . 3** Use the graph to find the wavelength of the wave. **[1 mark]**

**0 1 . 4** Write down the equation that links wave speed, frequency and wavelength. **[1 mark]**

**0 1 . 5** Calculate the frequency of the wave. **[3 marks]**

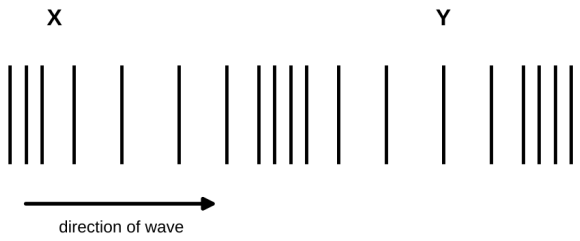
**Answer =** _____ Hz

**0 1 . 6** Calculate the time period of the wave. **[3 marks]**

**Answer =** _____ s

**0 1 . 7** State what is meant by the **frequency** of a wave. **[1 mark]**

**0 2** The diagram shows a longitudinal wave travelling along a spring.




**0 2 . 1** Name the parts of the wave labelled **X** and **Y**. **[2 marks]**

**0 2 . 2** Describe the difference between a transverse wave and a longitudinal wave. **[2 marks]**

**0 2 . 3** Give **one** example of a transverse wave and **one** example of a longitudinal wave. **[2 marks]**

**0 2 . 4** Waves transfer energy but not matter. Describe how you could show this using ripples on water. **[2 marks]**

**0 2 . 5** Explain why sound cannot travel through a vacuum. **[2 marks]**

**0 3** This question is about wave speed calculations.

**0 3 . 1** Ripples on water have a frequency of 5.0 Hz and a wavelength of 0.040 m. Calculate the speed of the ripples. **[3 marks]**

**Answer =** _____ m/s

**0 3 . 2** A sound wave travels through air at 340 m/s. It has a frequency of 256 Hz. Calculate the wavelength. **[3 marks]**

**Answer =** _____ m

**0 3 . 3** Radio waves travel at $3.0 \times 10^8$ m/s. A radio station broadcasts waves of wavelength 1500 m. Calculate the frequency. **[3 marks]**

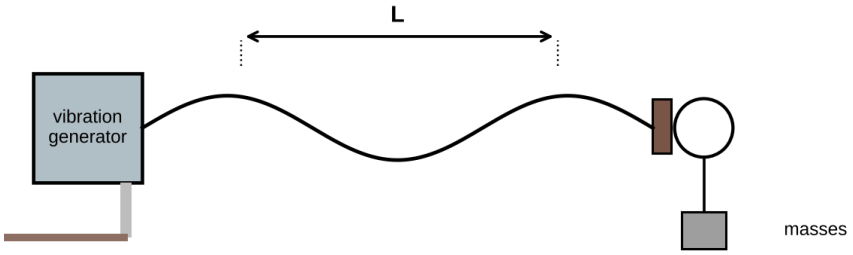
**Answer =** _____ Hz

**0 3 . 4** A ship sends a pulse of sound to the sea bed and the echo returns after 0.20 s. The speed of sound in sea water is 1500 m/s. Calculate the depth of the sea bed. **[4 marks]**

**Answer =** _____ m

**0 3 . 5** State what happens to the speed, frequency and wavelength of a wave when it moves from air into water. Assume the wave slows down. **[3 marks]**

**0 4** A student investigates waves on a stretched string. A vibration generator makes the string vibrate at a frequency of 100 Hz. The diagram shows the apparatus.




**0 4 . 1** The student measures the distance across 5 complete wavelengths as 0.60 m. Calculate the wavelength. **[2 marks]**

**Answer =** _____ m

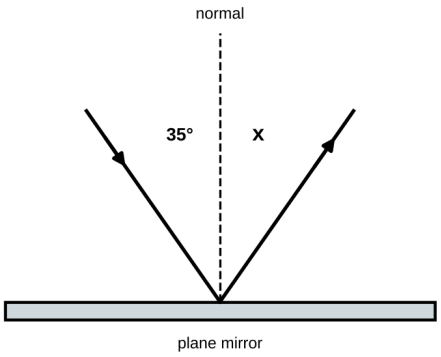
**0 4 . 2** Calculate the speed of the waves on the string. **[3 marks]**

**Answer =** _____ m/s

**0 4 . 3** Explain why the student measures the distance across 5 wavelengths instead of just one. **[2 marks]**

**0 4 . 4** Describe how the student could investigate how the wave speed depends on the tension in the string. **[3 marks]**

**0 5** A ray of light is reflected from a plane mirror, as shown in the diagram.




**0 5 . 1** Name the dashed line. **[1 mark]**

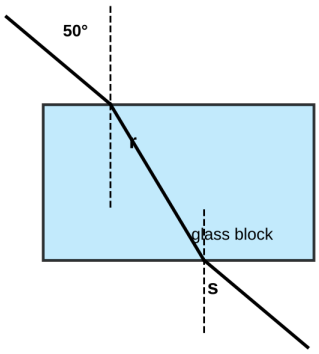
**0 5 . 2** State the size of angle **x**. **[1 mark]**

**0 5 . 3** State the law of reflection. **[1 mark]**

**0 5 . 4** Describe the difference between reflection from a smooth surface and from a rough surface. **[2 marks]**

**0 5 . 5** A student looks at an image of herself in a plane mirror. Describe **two** properties of the image. **[2 marks]**

**0 6** A ray of light passes through a rectangular glass block, as shown. The angle of incidence is 50°.




**0 6 . 1** State the name of the process that causes the ray to change direction as it enters the block. **[1 mark]**

**0 6 . 2** The angle of refraction **r** is 31°. Explain how the ray has changed direction when it enters the glass. **[2 marks]**

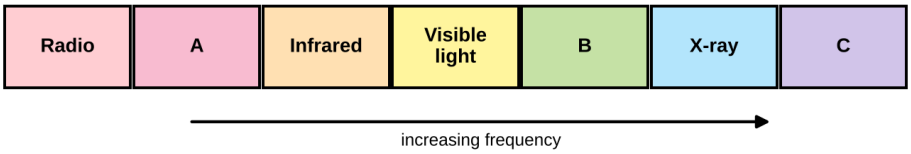
**0 6 . 3** State the size of angle **s**, where the ray leaves the block. **[1 mark]**

**0 6 . 4** Explain why the ray leaving the block is parallel to the incident ray. **[2 marks]**

**0 6 . 5** Explain, in terms of wave speed, why refraction happens. **[3 marks]**

**0 6 . 6** Describe how the student could use a ray box to find the relationship between angle of incidence and refraction. **[4 marks]**

**0 7** The diagram shows the electromagnetic spectrum. Some of the types of wave are labelled **A**, **B** and **C**.




**0 7 . 1** Name the types of wave **A**, **B** and **C**. **[3 marks]**

**0 7 . 2** Which type of wave in the diagram has the shortest wavelength? **[1 mark]**

**0 7 . 3** All electromagnetic waves travel at the same speed in a vacuum. State this speed. **[1 mark]**

**0 7 . 4** State **two** properties that are the same for all electromagnetic waves. **[2 marks]**

**0 7 . 5** Complete the table by giving one use of each type of electromagnetic wave. **[4 marks]**

| **Type of wave** | **Use** |
|---|---|
| Radio | |
| Microwaves | |
| Infrared | |
| X-rays | |

**0 8** This question is about the hazards and production of electromagnetic waves.

**0 8 . 1** Give **one** hazard of exposure to ultraviolet radiation. **[1 mark]**

**0 8 . 2** Gamma rays and X-rays are ionising radiation. Describe the possible effects of ionising radiation on living cells. **[2 marks]**

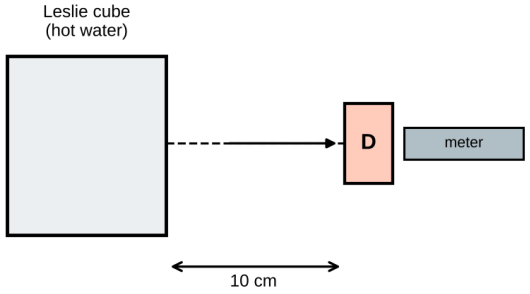
**0 8 . 3** A doctor uses X-rays to take an image of a broken arm. Explain why the doctor stands behind a lead screen. **[2 marks]**

**0 8 . 4** Explain how radio waves are produced and detected. **[3 marks]**

**0 8 . 5** Gamma rays come from changes in the nucleus. State what causes visible light and ultraviolet to be emitted from atoms. **[2 marks]**

**0 8 . 6** Suggest why microwaves can be used for cooking food. **[2 marks]**

**0 9** A student investigates infrared radiation emitted from different surfaces of a Leslie cube filled with hot water. The detector **D** is placed 10 cm from each face in turn. The table shows the results.




**0 9 . 1** Complete the table: rank the surfaces from best emitter (1) to worst (4). **[2 marks]**

| Surface | Detector reading | Rank |
|---|---|---|
| Matt black | 95 | |
| Shiny black | 62 | |
| Matt white | 58 | |
| Shiny silver | 14 | |

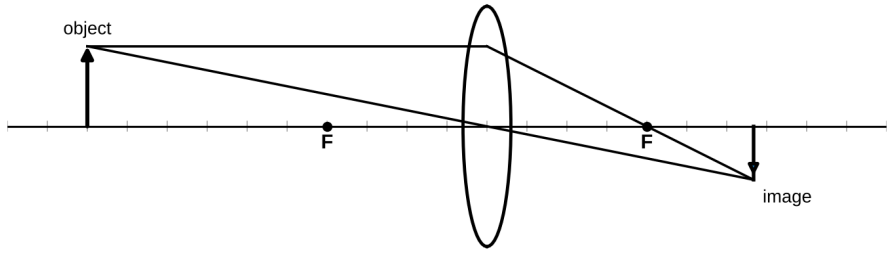
**0 9 . 2** State **two** variables that should be controlled. **[2 marks]**

**0 9 . 3** The student repeats the experiment with hot water in matt black and shiny silver cans. Predict which can cools more quickly and explain your prediction. **[3 marks]**

**0 9 . 4** State which surface is the best **absorber** of infrared radiation. **[1 mark]**

**0 9 . 5** Describe how the temperature of the Earth is affected by radiation from the Sun and from the Earth's surface. **[3 marks]**

**1 0** The diagram shows an object placed in front of a convex lens. **F** is the principal focus. The image is shown.




**1 0 . 1** Which two rays are drawn to find the position of the image? **[2 marks]**

**1 0 . 2** Describe the image. Choose from: real or virtual; upright or inverted; magnified or diminished. **[3 marks]**

**1 0 . 3** The object height is 2.0 cm and the image height is 1.3 cm. Calculate the magnification. **[3 marks]**

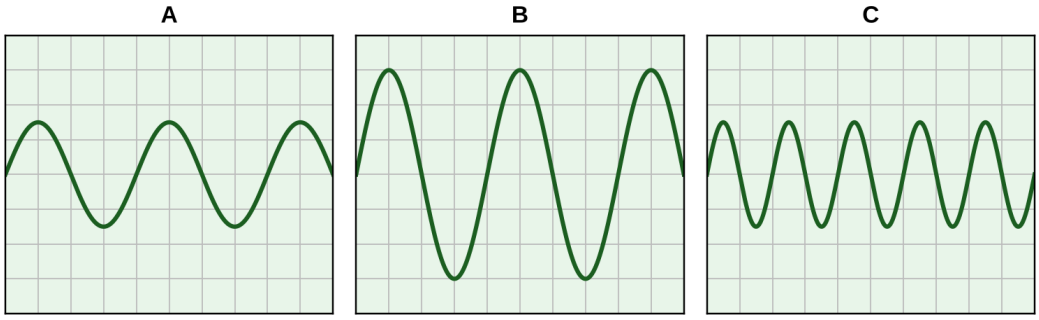
**Answer =** _____

**10.4** What is meant by a **real image**? **[1 mark]**

**10.5** An object is placed closer to a convex lens than the principal focus. Describe the image formed and give one use. **[3 marks]**

**10.6** State the effect of a concave lens on parallel rays of light. **[1 mark]**

**11** The oscilloscope traces **A**, **B** and **C** show three sounds. The time base is set to 0.50 ms per division.




**11.1** Which trace shows the loudest sound? Give a reason for your answer. **[2 marks]**

**11.2** Which trace shows the sound with the highest pitch? Give a reason. **[2 marks]**

**1 1 . 3** Trace A has a period of 4 divisions. Calculate the frequency of the sound for trace A. **[4 marks]**

**Answer =** _____ Hz

**1 1 . 4** State the range of human hearing. **[1 mark]**

**1 1 . 5** Describe how ultrasound is used for imaging an unborn baby. **[3 marks]**

**1 2** Earthquakes produce two types of seismic wave: P-waves and S-waves.

**1 2 . 1** State which type is longitudinal and which is transverse. **[2 marks]**

**1 2 . 2** S-waves cannot travel through the Earth's liquid outer core, but P-waves can. Explain how this provides evidence for the structure of the Earth. **[3 marks]**

**12.3** Explain why P-waves change direction as they pass through the Earth. **[2 marks]**

**13** Read the question and answer in full sentences.

**13.1** *Describe how you could use a ripple tank to measure the speed of water waves. Your answer should include the apparatus, the measurements taken and how the speed is calculated. **[6 marks]**

**13.2** *Evaluate the benefits and risks of using X-rays and ultraviolet radiation. Include one use of each. **[6 marks]**

**END OF QUESTIONS**

# MARK SCHEME – Topic 4.6: Waves

**How to use this mark scheme:** mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 01.1 | P: crest / peak ;<br>Q: trough ;<br>R: rest position / equilibrium ; | | 3 | AO1<br>4.6.1.1 |
| 01.2 | 3.0 (cm) | | 1 | AO3<br>4.6.1.2 |
| 01.3 | 0.80 (m) | | 1 | AO3<br>4.6.1.2 |
| 01.4 | wave speed = frequency × wavelength | | 1 | AO1<br>4.6.1.2 |
| 01.5 | f = v ÷ λ ;<br>= 2.0 ÷ 0.80 ;<br>2.5 (Hz) ; | | 3 | AO2<br>4.6.1.2 |
| 01.6 | T = 1 ÷ f ;<br>1 ÷ 2.5 ;<br>0.40 (s) ; | Allow ecf from part 5. | 3 | AO2<br>4.6.1.2 |
| 01.7 | the number of waves passing a point each second / number of oscillations per second | | 1 | AO1<br>4.6.1.2 |
| 02.1 | X: compression ;<br>Y: rarefaction ; | | 2 | AO1<br>4.6.1.1 |
| 02.2 | transverse: oscillations are perpendicular (at right angles) to the direction of energy transfer ;<br>longitudinal: oscillations are parallel to the direction of energy transfer ; | | 2 | AO1<br>4.6.1.1 |
| 02.3 | transverse: light / any EM wave / ripples / wave on rope ;<br>longitudinal: sound / ultrasound / P-waves ; | | 2 | AO1<br>4.6.1.1 |
| 02.4 | a floating cork / ball bobs up and down ;<br>does not move along with the wave ; | | 2 | AO2<br>4.6.1.1 |
| 02.5 | sound waves are vibrations of particles ;<br>no particles in a vacuum to pass on the vibrations ; | | 2 | AO2<br>4.6.1.1 |
| 03.1 | v = f λ ;<br>= 5.0 × 0.040 ;<br>0.20 (m/s) ; | | 3 | AO2<br>4.6.1.2 |
| 03.2 | λ = v ÷ f ;<br>= 340 ÷ 256 ;<br>1.3 (m) ; | Allow 1.33. | 3 | AO2<br>4.6.1.2 |
| 03.3 | f = v ÷ λ ;<br>= $3.0 \times 10^8 \div 1500$ ;<br>$2.0 \times 10^5$ (Hz) ; | | 3 | AO2<br>4.6.2.1 |

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 03.4 | distance = speed × time = 1500 × 0.20 ;<br>= 300 m ;<br>÷ 2 for there and back ;<br>150 (m) ; | | 4 | AO3<br>4.6.1.2 |
| 03.5 | speed decreases ;<br>frequency stays the same ;<br>wavelength decreases ; | | 3 | AO2<br>4.6.1.2 |
| 04.1 | 0.60 ÷ 5 ;<br>0.12 (m) ; | | 2 | AO2<br>4.6.1.2 RP |
| 04.2 | v = f λ ;<br>= 100 × 0.12 ;<br>12 (m/s) ; | | 3 | AO2<br>4.6.1.2 RP |
| 04.3 | reduces the percentage uncertainty / error ;<br>easier to measure a larger distance accurately ; | | 2 | AO3<br>4.6.1.2 RP |
| 04.4 | change the tension by changing the masses on the end of the string ;<br>measure wavelength each time at the same frequency and calculate speed ;<br>plot speed against tension / repeat for more readings ; | | 3 | AO3<br>4.6.1.2 RP |
| 05.1 | normal | | 1 | AO1<br>4.6.2.2 |
| 05.2 | 35° | | 1 | AO3<br>4.6.2.2 |
| 05.3 | angle of incidence = angle of reflection | | 1 | AO1<br>4.6.2.2 |
| 05.4 | smooth: specular reflection – parallel rays stay parallel / clear image ;<br>rough: diffuse reflection – rays scattered in many directions ; | | 2 | AO1<br>4.6.2.2 |
| 05.5 | virtual ;<br>same size as the object / upright / laterally inverted / same distance behind the mirror ; | Any two. | 2 | AO1<br>4.6.2.2 |
| 06.1 | refraction | | 1 | AO1<br>4.6.2.2 |
| 06.2 | bends towards the normal ;<br>because it slows down (glass is optically denser than air) ; | | 2 | AO2<br>4.6.2.2 |
| 06.3 | 50° | | 1 | AO3<br>4.6.2.2 |
| 06.4 | it bends away from the normal on leaving ;<br>by the same amount as it bent towards the normal on entering (the sides are parallel) ; | | 2 | AO3<br>4.6.2.2 |
| 06.5 | the speed of the wave changes when it enters the new medium ;<br>one side of the wavefront slows down first ;<br>so the wavefront (ray) changes direction ; | | 3 | AO2<br>4.6.2.2 |

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 06.6 | place the block on paper and draw round it ;<br>shine a ray at the block and mark the incident, emergent rays ;<br>draw the normal and join marks to find the refracted ray ;<br>measure angles with a protractor and repeat for different incident angles ; | | 4 | AO3<br>4.6.2.2 RP |
| 07.1 | A: microwaves ;<br>B: ultraviolet ;<br>C: gamma (rays) ; | | 3 | AO1<br>4.6.2.1 |
| 07.2 | C / gamma | | 1 | AO1<br>4.6.2.1 |
| 07.3 | $3.0 \times 10^8$ (m/s) | | 1 | AO1<br>4.6.2.1 |
| 07.4 | transverse ;<br>travel at the same speed in a vacuum / transfer energy ; | | 2 | AO1<br>4.6.2.1 |
| 07.5 | radio: communication / broadcasting TV, radio ;<br>microwaves: cooking / satellite communication ;<br>infrared: thermal imaging / remote controls / heaters ;<br>X-rays: medical imaging of bones / airport security ; | | 4 | AO1<br>4.6.2.1 |
| 08.1 | skin cancer / eye damage / sunburn / premature ageing | | 1 | AO1<br>4.6.2.1 |
| 08.2 | can damage / mutate DNA ;<br>can cause cancer / cell death ; | | 2 | AO1<br>4.6.2.1 |
| 08.3 | X-rays are absorbed by lead ;<br>to protect from repeated exposure (cancer risk) ; | | 2 | AO3<br>4.6.2.1 |
| 08.4 | produced by oscillations / alternating current in an electrical circuit ;<br>radio waves are absorbed by a receiver (aerial) ;<br>causing an alternating current in the receiver circuit with the same frequency ; | | 3 | AO1<br>4.6.2.1 |
| 08.5 | electrons change energy level ;<br>moving from higher to lower energy level / absorbing and emitting ; | | 2 | AO1<br>4.6.2.1 |
| 08.6 | microwaves are absorbed by water molecules in food ;<br>making them vibrate / transferring energy to the thermal store ; | | 2 | AO2<br>4.6.2.1 |
| 09.1 | matt black 1 ;<br>shiny silver 4 (and matt white 3, shiny black 2) ; | | 2 | AO3<br>4.6.3.2 RP |
| 09.2 | distance from the cube to the detector ;<br>temperature of the water / same detector / same angle ; | | 2 | AO3<br>4.6.3.2 RP |
| 09.3 | matt black can ;<br>better emitter of infrared radiation ;<br>so transfers energy to the surroundings faster ; | | 3 | AO3<br>4.6.3.2 |

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 09.4 | matt black | | 1 | AO1<br>4.6.3.2 |
| 09.5 | the Earth absorbs radiation from the Sun (energy in) ;<br>and emits infrared radiation (energy out) ;<br>temperature is constant when energy in = energy out ; | | 3 | AO2<br>4.6.3.2 |
| 10.1 | a ray parallel to the axis, refracted through the principal focus ;<br>a ray through the centre of the lens, which does not bend ; | | 2 | AO2<br>4.6.2.3 |
| 10.2 | real ;<br>inverted ;<br>diminished ; | | 3 | AO2<br>4.6.2.3 |
| 10.3 | magnification = image height ÷ object height ;<br>1.3 ÷ 2.0 ;<br>0.65 ; | Allow 0.67 for 1.33 cm. | 3 | AO2<br>4.6.2.3 |
| 10.4 | an image formed where light rays actually meet / can be projected on a screen | | 1 | AO1<br>4.6.2.3 |
| 10.5 | virtual ;<br>upright and magnified ;<br>magnifying glass ; | | 3 | AO2<br>4.6.2.3 |
| 10.6 | rays diverge / spread out | | 1 | AO1<br>4.6.2.3 |
| 11.1 | B ;<br>largest amplitude ; | | 2 | AO3<br>4.6.1.1 |
| 11.2 | C ;<br>highest frequency / most waves in the same time ; | | 2 | AO3<br>4.6.1.1 |
| 11.3 | period = 4 × 0.50 = 2.0 ms ;<br>= 0.0020 s ;<br>f = 1 ÷ T ;<br>500 (Hz) ; | | 4 | AO3<br>4.6.1.2 |
| 11.4 | 20 Hz to 20 000 Hz | | 1 | AO1<br>4.6.2.1 |
| 11.5 | ultrasound waves are sent into the body ;<br>waves are partly reflected at boundaries between tissues ;<br>reflections detected and processed to give an image ; | | 3 | AO1<br>4.6.3.1 |
| 12.1 | P-waves are longitudinal ;<br>S-waves are transverse ; | | 2 | AO1<br>4.6.3.1 |
| 12.2 | S-waves are not detected on the opposite side of the Earth ;<br>transverse waves cannot pass through a liquid ;<br>so part of the core is liquid ; | | 3 | AO3<br>4.6.3.1 |
| 12.3 | density of the Earth changes with depth ;<br>so the wave speed changes and the waves refract ; | | 2 | AO2<br>4.6.3.1 |

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 13.1 | **Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together.<br>**Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places.<br>**Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning.<br>**0 marks**: No relevant content.<br>**Indicative content** (credit any other correct science):<br>• Level 3 (5–6): clear, logical method including a measure of frequency and wavelength and use of $v = f \lambda$.<br>• Level 2 (3–4): some measurements and calculation described.<br>• Level 1 (1–2): simple statements.<br>• Indicative content: ripple tank with motor / dipper and lamp; image projected on screen or paper; use a strobe light to freeze the pattern (or video); measure distance across several wavelengths with a ruler and divide; scale factor for image; frequency from the generator or by counting waves passing a point in a time; calculate $v = f \times \lambda$; repeat and calculate a mean. | | 6 | AO3<br>4.6.1.2 RP |
| 13.2 | **Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together.<br>**Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places.<br>**Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning.<br>**0 marks**: No relevant content.<br>**Indicative content** (credit any other correct science):<br>• Level 3 (5–6): balanced evaluation of both radiations with uses, hazards and precautions and a judgement.<br>• Level 2 (3–4): some benefits and risks for both.<br>• Level 1 (1–2): simple statements.<br>• Indicative content: X-rays – imaging bones, detecting broken bones, cancer treatment (radiotherapy); high frequency, ionising, can cause mutations / cancer; precautions: lead aprons, short exposure, radiographers behind screens; UV – sun tanning, sterilising water, security marks; can cause skin cancer, eye damage; precautions: sunscreen, sunglasses, limit time; benefits outweigh risks if controlled. | | 6 | AO3<br>4.6.2.1 |

# Where marks are lost in Topic 4.6

*Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.*

- ☐ **Learn the wave labels:** amplitude is the **maximum** displacement from the rest position (not peak to trough); wavelength is crest to crest.
- ☐ **Wave equations:** $v = f \lambda$, $T = 1 \div f$. Check units (Hz, m, m/s). Convert cm and ms before calculating.
- ☐ **Reflection:** angles are measured from the **normal**, not from the surface.
- ☐ **Refraction:** into a denser material the ray bends **towards** the normal (it slows down); out of it bends **away**. The frequency stays the same; speed and wavelength change.
- ☐ **EM spectrum order:** radio, microwave, infrared, visible, ultraviolet, X-ray, gamma — increasing frequency, decreasing wavelength. All travel at $3.0 \times 10^8$ m/s in a vacuum.
- ☐ **Ionising radiation** (UV, X-rays, gamma) damages cells and can cause cancer. Say what damage and which precaution.
- ☐ **Lenses:** for the ray diagram use 2 rays — parallel ray through F, and a straight ray through the centre. State real/virtual, upright/inverted, magnified/diminished.
- ☐ **Sound:** louder = greater amplitude; higher pitch = higher frequency. Sound is longitudinal and needs a medium.
- ☐ **Infrared practical:** matt black surfaces are the best emitters and the best absorbers; shiny silver the worst. Keep distance and temperature constant.